# Louis Nirenberg

Terence Tao · 27 January 2020

https://terrytao.wordpress.com/2020/01/27/louis-nirenberg/

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I just heard the news that Louis Nirenberg died a few days ago, aged 94. Nirenberg made a vast number of contributions to analysis and PDE (and his work has come up repeatedly on my own blog); I wrote about his beautiful moving planes argument with Gidas and Ni to establish symmetry of ground states in this post on the occasion of him receiving the Chern medal, and on how his extremely useful interpolation inequality with Gagliardo (generalising a previous inequality of Ladyzhenskaya) can be viewed as an amplification of the usual Sobolev inequality in this post. Another fundamentally useful inequality of Nirenberg is the John-Nirenberg inequality established with Fritz John: if a (locally integrable) function $f : \mathbf{R} \to \mathbf{R}$ (which for simplicity of exposition we place in one dimension) obeys the bounded mean oscillation property

$$\frac{1}{|I|} \int_I |f(x) - f_I|\, dx \leq A \tag{1}$$

for all intervals $I$, where $f_I := \frac{1}{|I|} \int_I f$ is the average value of $f$ on $I$, then one has exponentially good large deviation estimates

$$\frac{1}{|I|} |\{x \in I : |f(x) - f_I| \geq \lambda A\}| \leq \exp(-c\lambda) \tag{2}$$

for all $\lambda > 0$ and some absolute constant $c$. This can be compared with Markov's inequality, which only gives the far weaker decay

$$\frac{1}{|I|} |\{x \in I : |f(x) - f_I| \geq \lambda A\}| \leq \frac{1}{\lambda}. \tag{3}$$

The point is that (1) is assumed to hold not just for a given interval $I$, but also all subintervals of $I$, and this is a much more powerful hypothesis, allowing one for instance to use the standard Calderon-Zygmund technique of stopping time arguments to "amplify" (3) to (2). Basically, for any given interval $I$, one can use (1) and repeated halving of the interval $I$ until significant deviation from the mean is encountered to locate some disjoint exceptional subintervals $J$ where $f_J$ deviates from $f_I$ by $O(A)$, with the total measure of the $J$ being a small fraction of that of $I$ (thanks to a variant of (3)), and with $f$ staying within $O(A)$ of $f_I$ at almost every point of $I$ outside of these exceptional intervals. One can then establish (2) by an induction on $\lambda$. (There are other proofs of this inequality also, e.g., one can use Bellman functions, as discussed in this old set of notes of mine.) Informally, the John-Nirenberg inequality asserts that functions of bounded mean oscillation are "almost as good" as bounded functions, in that they almost always stay within a bounded distance from their mean, and in fact the space BMO of functions of bounded mean oscillation ends up being superior to the space $L^\infty$ of bounded measurable functions for many harmonic analysis purposes (among other things, the space is more stable with respect to singular integral operators).

I met Louis a few times in my career; even in his later years when he was wheelchair-bound, he would often come to conferences and talks, and ask very insightful questions at the end of the lecture (even when it looked like he was asleep during much of the actual talk!). I have a vague memory of him asking me some questions in one of the early talks I gave as a postdoc; I unfortunately do not remember exactly what the topic was (some sort of PDE, I think), but I was struck by how kindly the questions were posed, and how patiently he would listen to my excited chattering about my own work.

## Comments (15)

**Steven Heilman** · 27 January 2020

He brought his Chern medal in a nondescript plastic bag, to show students at the celebration honoring that achievement. Perhaps greatness within a humble exterior best describes him. Very sad news. He will be missed.

**Anonymous** · 27 January 2020

Awesome mathematician.Deepest respects to him.I wish i could have been a little older so as to learn from the best.

**thepoliblog** · 27 January 2020

Hans Bethe, too, had the ability to ask penetrating questions after appearing to doze through a talk.

**ahmedalshabi** · 27 January 2020

RIP

**David W. Fry** · 28 January 2020

Dear Dr. Tao,
I am extremely disheartened to hear about your colleague. It is obvious that you had a great deal of respect and admiration for Dr. Nirenberg.
Very Sincerely,
David

**Brenton LeMesurier** · 29 January 2020

One fond memory from my days as a student at the Courant Institute was the gentle questions that Nirenberg would ask in seminars, not because he did not know the answer, but to ensure that us graduate students got the explanation we needed.

**Kirill Mayorov** · 29 January 2020

Needless to say how many bright disciples he produced: https://www.genealogy.math.ndsu.nodak.edu/id.php?id=13410
In particular, Walter Craig (https://math.mcmaster.ca/news-events/news/1919-walter-craig-1953-2019.html) was one of them.

> **Sophie MacDonald** · 1 February 2020
>
> Walter was my first research advisor, for an undergraduate summer project. The day that I met with him to decide the topic, there was an elderly man visiting him as well, who in retrospect looked a lot like Nirenberg — at the time, I had no idea who he was. Walter was exceptionally generous to me that summer and in the few remaining years of his life. Gone far too soon.

**a** · 30 January 2020

Is the JN inequality applicable to probability in special formalized cases?

> **Terence Tao** · 30 January 2020
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> There is a certain amount of literature on martingales of bounded mean oscillation (see e.g., this text of Petersen), in which I would imagine the John-Nirenberg inequality plays a useful role. However I am not familiar with the details.

**Anonymous** · 1 February 2020

Dear Sir , Pro.Tao

Today , I would like to have a suggestion for you after I have thoughts for many days.

Pro.Tao, Maths is very important in our life. Now, Biology and Chemistry are more important. You can stop do maths in one month in order to focus on other field. ( Although it is not your field, I know well). But you use your intelligent ( IQ 230) which God presented to you to save many people in the world. I mean you can find a medicine solution in your dream at night to kill virus corona. Nothing is impossible! I wait a miracle from you. If this becomes true, all human beings in the world would be grateful to you deeply. You become a great hero , a living saint for mankind. After that , you certainly get a Nobel prize in October , 2020.

Thank you, Sir.

**VCP** · 1 February 2020

Touching obituary of an impactful mathematician, who will certainly be remembered for many reasons.

(I wonder whether in the definition of $f_I$ after (1), the averaging term $\int_I f(x)dx$ is missing.)

*[Corrected, thanks – T.]*

**Anonymous** · 1 February 2020

Touching obituary but you forgot to mention the other mathematical great that died — Koebe

**Various | Not Even Wrong** · 3 February 2020

[…] I was sorry to hear recently about the death of mathematician Louis Nirenberg. Kenneth Chang at the New York Times has written an excellent obituary. Terry Tao has some comments here. […]

**Leo's blog · My change to WP** · 9 February 2020

[…] WordPress (henceforth, WP), I decided to give it a shot (see, for example, Andrew Gelman and Terence Tao. I actually “stole” the style from the […]